# Explanation of the provided code

focuses on three components:

**Collateral Rating**

**Financial Rating**

**Bank Profitability (Yield)**

## 1] Collateral Rating

This part evaluates the quality of assets (e.g., cash, property) a borrower offers to secure the loan.

- **What it does**: Rates assets based on how easy they are to sell, how stable their value is, and if they have legal issues.
- **Functions**:
  - assign_collateral_rating: Gives a score (1-5) for each asset based on:
    - **Liquidity**: Can it be turned into cash quickly? (High=5, Medium=3, Low=1)
    - **Stability**: Does the value stay steady? (Stable=5, Moderate=3, Volatile=1)
    - **Legal Clarity**: Are there ownership problems? (Clear=5, Minor Issues=3, Issues=1)
    - Combines these using weights (50% liquidity, 30% stability, 20% legal clarity) and rounds to a 1-5 score (5=Outstanding, 1=Poor).
  - calculate_collateral_score: Averages ratings of multiple assets (e.g., cash, property) using weights (e.g., 40% cash, 30% property).
  - calibrate_collateral: Converts the average score to a 1-4 scale (4=Excellent, 3=Good, 2=Medium, 1=Poor) for decision-making.
- **Why it matters**: Ensures the bank can recover money if the borrower doesn't repay by checking asset quality.

## 2] Financial Rating

This part checks the borrower's financial health to see if they can repay the loan.

- **What it does**: Scores financial factors like net worth or debt to assess the borrower's stability.
- **Functions**:
  - calculate_financial_score: Averages ratings (1-5) of financial factors (e.g., Net Worth=5, Debt=3) using weights (e.g., 30% Net Worth, 20% Debt).
  - calibrate_financial: Converts the average score to a 1-4 scale (4=Excellent, 3=Good, 2=Medium, 1=Bad).
- **Why it matters**: Helps the bank know if the borrower is financially strong enough to pay back the loan.

**<u>3] Bank Profitability (Yield)</u>**

This part rates how profitable the loan is for the bank.

- **What it does**: Assigns a score (1-3) to the loan's profitability based on its interest rate or return.
- **Function**:
  - assign_yield_rating: Takes a direct input for profitability (3=Excellent, 2=Reasonable, 1=Poor). Defaults to 1 if no input is given.
- **Why it matters**: Ensures the loan makes money for the bank, balancing risk and reward.